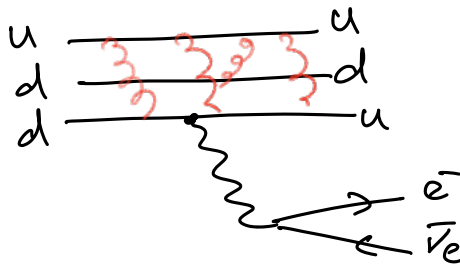


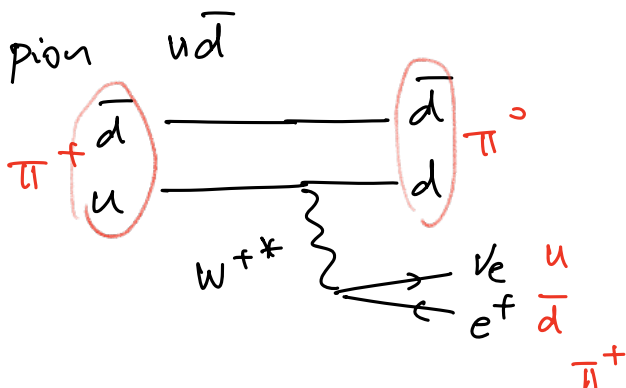
Weak interaction of quarks



$$n \rightarrow p e^- \bar{\nu}_e$$

β decay of pions

spectator diagram



$$Q = m_{\pi^+} - m_{\pi^0} - m_e > 0 \quad \pi^+ \rightarrow \pi^0 e^+ \nu_e$$

$$\pi^+ \rightarrow \pi^0 \pi^+ \quad Q < 0$$

π^+ DECAY MODES

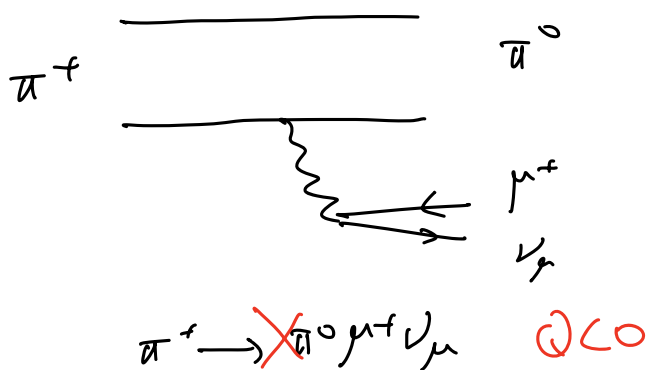
π^- modes are charge conjugates of the modes below.

For decay limits to particles which are not established, see the section on Searches for Axions and Other Very Light Bosons.

Mode	Fraction (Γ_i/Γ)	Confidence level
$\Gamma_1 \quad \mu^+ \nu_\mu$	[a] (99.98770 $\pm$ 0.00004) %	
$\Gamma_2 \quad \mu^+ \nu_\mu \gamma$	[b] (2.00 $\pm$ 0.25) $\times 10^{-4}$	
$\Gamma_3 \quad e^+ \nu_e$	[a] (1.230 $\pm$ 0.004) $\times 10^{-4}$	
$\Gamma_4 \quad e^+ \nu_e \gamma$	[b] (7.39 $\pm$ 0.05) $\times 10^{-7}$	
$\Gamma_5 \quad e^+ \nu_e \pi^0$	(1.036 $\pm$ 0.006) $\times 10^{-8}$	
$\Gamma_6 \quad e^+ \nu_e e^+ e^-$	(3.2 $\pm$ 0.5) $\times 10^{-9}$	
$\Gamma_7 \quad \mu^+ \nu_\mu \nu \bar{\nu}$	< 9 $\times 10^{-6}$	90%
$\Gamma_8 \quad e^+ \nu_e \nu \bar{\nu}$	< 1.6 $\times 10^{-7}$	90%

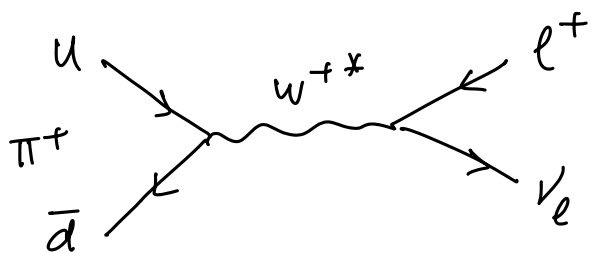
Lepton Family number (LF) or Lepton number (L) violating modes

$\Gamma_9 \quad \mu^+ \bar{\nu}_e$	L	[c] < 1.5	$\times 10^{-3}$	90%
$\Gamma_{10} \quad \mu^+ \nu_e$	LF	[c] < 8.0	$\times 10^{-3}$	90%
$\Gamma_{11} \quad \mu^- e^+ e^+ \nu$	LF	< 1.6	$\times 10^{-6}$	90%



S-channel

annihilation channel



$$Q = m_\pi - m_e - m_\nu > 0 \quad l = e, \mu$$

$$m_\pi = 140 \text{ MeV}$$

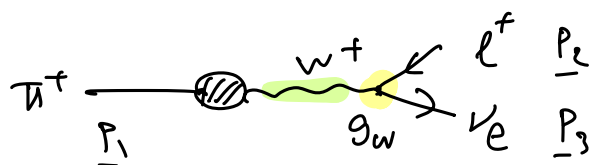
$$m_\mu = 106 \text{ MeV}$$

$$m_e = 0.5 \text{ MeV}$$

$$Q_e > Q_\mu$$

why $\text{BF}(\pi \rightarrow \mu) \gg \text{BF}(\pi \rightarrow e)$?

pion decay constant



$$\mathcal{M} \sim \frac{g_w}{q^2 - m_w^2} (\bar{u}(p_3) \gamma^\mu (1 - \gamma^5) u(p_2)) g_w \bar{u}(p_1)$$

$$q^2 \approx m_\pi^2 \ll m_w^2$$

4-momentum of pion.

$S=0$ for pion

$$\Gamma_\pi \propto \langle |M|^2 \rangle \times \rho(\text{phase space})$$

init state $s=0$

$$\langle |M|^2 \rangle = \sum_{\text{final state spins}} \sum_{\text{average over init. state spins}} |M|^2$$

Final state

$$e^+ \xrightarrow{=} \xrightarrow{=} \nu_e$$

$$\langle |M|^2 \rangle = \left(\frac{g_W}{2m_W} \right)^4 f_\pi^2 m_e^2 (m_\pi^2 - m_e^2)$$

In rest frame of π^+

$$e \leftarrow \bullet \rightarrow \nu_e \quad \vec{p} = \vec{p}_e + \vec{p}_\nu$$

$$\vec{p}_e = -\vec{p}_\nu \equiv \vec{p}_2 \quad p_2 \equiv |\vec{p}_2|$$

en. conservation

$$m_\pi = E_e + E_\nu = \sqrt{m_e^2 + p_2^2} + p_2$$

$$p_2 = \frac{1}{2m_\pi} (m_\pi^2 - m_e^2)$$

$$\Gamma_\pi = \frac{1}{8\pi} \frac{|\vec{p}_2|}{m_\pi^2} \langle |M|^2 \rangle \quad \text{compute } \Gamma(\pi \rightarrow \mu), \Gamma(\pi \rightarrow e)$$

$$\frac{\Gamma_e}{\Gamma_\mu} = \frac{\Gamma(\pi^+ \rightarrow e^+ \nu_e)}{\Gamma(\pi^+ \rightarrow \mu^+ \nu_\mu)} = \frac{m_e^2}{m_\mu^2} \underbrace{\frac{(m_\pi^2 - m_e^2)}{(m_\pi^2 - m_\mu^2)}}_{\langle |M|^2 \rangle} \underbrace{\frac{(m_\pi^2 - m_e^2)}{(m_\pi^2 - m_\mu^2)}}_{\text{phase space } |\vec{p}_2|} \frac{2m_\pi^3}{2m_\pi^3}$$

$$= \left(\frac{m_e}{m_\mu} \right)^2 \frac{(m_\pi^2 - m_e^2)^2}{(m_\pi^2 - m_\mu^2)^2}$$

$$\left(\frac{m_e}{m_\mu} \right)^2 = \left(\frac{0.5}{106} \right)^2 \approx 10^{-4} \ll 1$$

$$\frac{\Gamma_e}{\Gamma_\mu} = 1.23 \times 10^{-4} \quad \text{in agreement with exp. data}$$

why suppression of $\pi^+ \rightarrow e^+ \nu_e$

$$e^+ \xleftarrow{\Rightarrow \pi^0} \nu_e \quad S=0$$

$$S_z = 0$$

$$e^+ \xleftarrow{\Rightarrow \pi^0} \nu_e$$

$$S_z = -1$$

$$\beta = \frac{p_e}{E_e}$$

$$E_e \approx \frac{m_\pi}{2} = 70 \text{ MeV}$$

$$\beta \approx \frac{p}{m} \approx \frac{70}{0.5} = 140$$

$$\Rightarrow \beta \approx 1$$

$\Rightarrow e^+$ should be right handed.

$$\mu^+ \xleftarrow{\Rightarrow} \nu_\mu$$

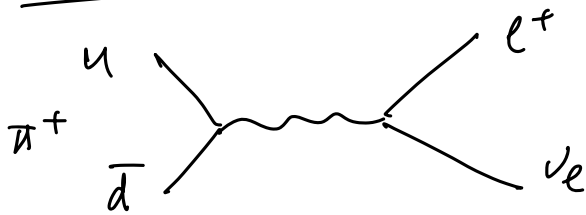
$$\beta_\mu = \frac{p_\mu}{E_\mu}$$

$$\beta_\mu = \frac{p_\mu}{m_\mu} = \frac{m_\pi/2}{m_\mu} \approx 0.5$$

$$\Rightarrow \beta < 1$$

$\Rightarrow \mu^+$ can be both right-handed and left-handed.

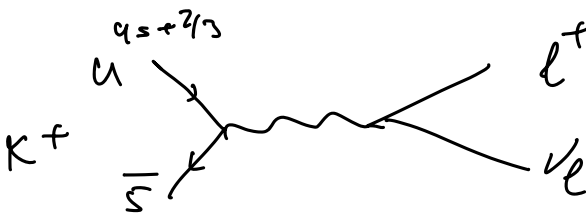
Compare d and s quarks



$$\pi^+ \rightarrow \mu^+ \nu_\mu$$

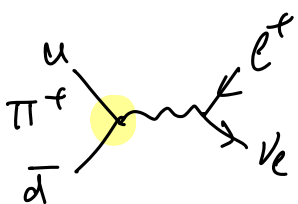
$$M_\pi = 140 \text{ MeV}$$

$$M_\mu = 106 \text{ MeV}$$



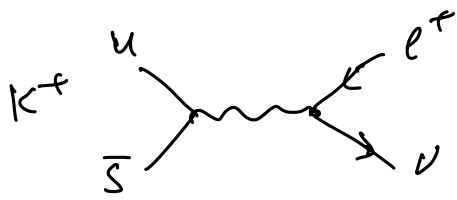
$$K^+ \rightarrow \mu^+ \nu_\mu$$

$$M_K = 500 \text{ MeV}$$



$$\mu \sim f_\pi \frac{g_W^2}{m_W^2} m_l^2 (m_\pi^2 - m_l^2)$$

$$\Gamma(\pi \rightarrow \mu) \sim G_F^2 \frac{m_l^2}{m_\pi^3} (m_\pi^2 - m_l^2)^2 f_\pi^2$$



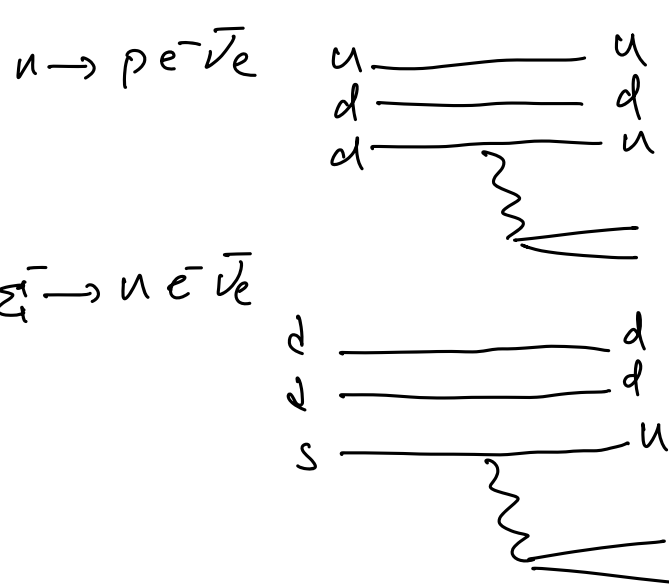
$$\Gamma(K \rightarrow \mu \nu) \sim G_{F_K}^2 \frac{m_e^2}{m_K^3} (m_K^2 - m_e^2)^2 f_K^2$$

$$\frac{\Gamma(K^+ \rightarrow \mu^+ \nu_\mu)}{\Gamma(\pi^+ \rightarrow \mu^+ \nu_\mu)} = \left(\frac{G_{F_K}}{G_{F_\pi}} \right)^2 \left(\frac{f_K}{f_\pi} \right)^2 \left(\frac{m_\pi}{m_K} \right)^3 \frac{(m_K^2 - m_\mu^2)^2}{(m_\pi^2 - m_\mu^2)^2}$$

$$\frac{\Gamma(K^+ \rightarrow \mu^+ \nu_\mu)}{\Gamma(\pi^+ \rightarrow \mu^+ \nu_\mu)} = \frac{BF(K^+ \rightarrow \mu^+ \nu)}{BF(\pi^+ \rightarrow \mu^+ \nu)} \frac{\Gamma_{tot}^K}{\Gamma_{tot}^\pi} \rightarrow \frac{\tau_\pi}{\tau_K}$$

$$BF(K^+ \rightarrow \mu^+ \nu) = \frac{\# \text{ decays with } \mu}{\# \text{ total } K \text{ decays}}$$

$\Rightarrow \left(\frac{G_{F_K}}{G_{F_\pi}} \right)^2 \simeq 0.05 \Rightarrow$ s different than d coupling
 $\Rightarrow$ β decay different for K^+ , π^+ ? ?



n DECAY MODES

	Mode	Fraction (Γ_i/Γ)	Confidence level
Γ_1	$p e^- \bar{\nu}_e$	100	%
Γ_2	$p e^- \bar{\nu}_e \gamma$	[a] $(9.2 \pm 0.7) \times 10^{-3}$	
Γ_3	hydrogen-atom $\bar{\nu}_e$	< 2.7	$\times 10^{-3}$ 95%

Σ^- DECAY MODES

	Mode	Fraction (Γ_i/Γ)	Confidence level
Γ_1	$n \pi^-$	$(99.848 \pm 0.005) \%$	
Γ_2	$n \pi^- \gamma$	[a] $(4.6 \pm 0.6) \times 10^{-4}$	
Γ_3	$n e^- \bar{\nu}_e$	$(1.017 \pm 0.034) \times 10^{-3}$	
Γ_4	$n \mu^- \bar{\nu}_\mu$	$(4.5 \pm 0.4) \times 10^{-4}$	
Γ_5	$\Lambda e^- \bar{\nu}_e$	$(5.73 \pm 0.27) \times 10^{-5}$	
Γ_6	$\Sigma^+ X$	< 1.2	$\times 10^{-4}$ 90%

$$K^+ \rightarrow \mu \nu \quad \Delta S = 1$$

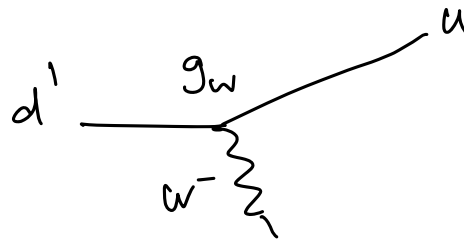
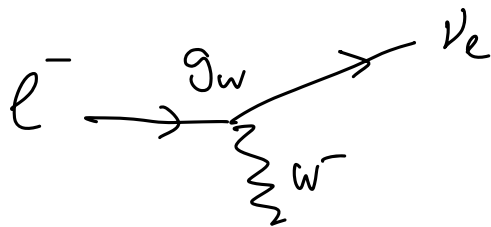
$$\pi^+ \rightarrow \mu \nu \quad \Delta S = 0$$

$$\Sigma^- \rightarrow n e^- \bar{\nu}_e$$

$$n \rightarrow p e^- \bar{\nu}_e$$

$\Gamma(\Delta S = 1) \ll \Gamma(\Delta S = 0)$ Exp. evidence.

Solution: Cabibbo Angle



W^- couples to $\begin{pmatrix} l^- \\ \nu_e \end{pmatrix}$
with g_w

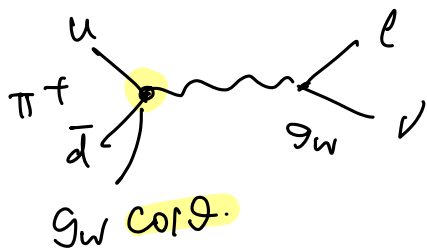
$$d' = d \cdot \cos\theta + s \cdot \sin\theta$$

θ : Cabibbo angle (mixing between d and s)

weak interaction basis $\neq$

Flavor basis in QCD.

$$d' \xrightarrow{g_w} u = d \cdot \cos\theta \xrightarrow{g_w} u + s \cdot \sin\theta \xrightarrow{g_w} u$$



$$\Gamma \propto G_F^2 \cos^2\theta m_\mu^2 (m_\pi^2 - m_\mu^2)^2 \frac{f_\pi^2}{m_\pi^3}$$

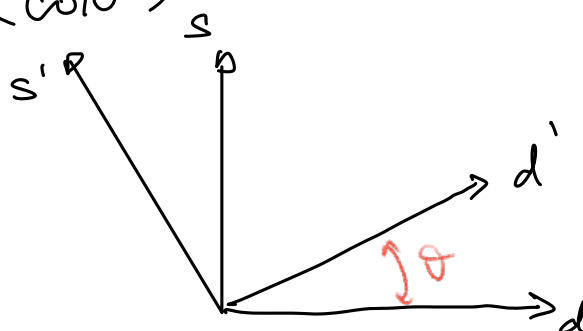
$$G_F = \frac{g_w^2}{m_W^2}$$



$$\Gamma \propto G_F^2 \sin^2\theta m_\mu^2 (m_K^2 - m_\mu^2)^2 \frac{f_K^2}{m_K^3}$$

$$\frac{\Gamma(K^+ \rightarrow \mu\nu)}{\Gamma(\pi^+ \rightarrow \mu\nu)} = (\tan\theta)^2 \underbrace{F(m_\mu, m_\pi, m_K, f_\pi, f_K)}_{\text{algebraic factor}}$$

$$\Rightarrow \left(\frac{\sin\theta}{\cos\theta} \right)^2 \leq 0.05 \Rightarrow \sin\theta \leq 0.25$$



weak interaction basis

Flavor eigenstates

$$\Gamma(\Delta S=0) \propto \cos^2 \theta \quad \Rightarrow \quad \text{Cabibbo angle explains all exp. data w/ } \Delta S=0$$

$$\Gamma(\Delta S=1) \propto \sin^2 \theta. \quad \Delta S=1$$

To determine θ , let us compare the rates for $K^+ \rightarrow \mu^+ + \nu$ and $\pi^+ \rightarrow \mu^+ + \nu$; we find

$$\Gamma(K^+ \rightarrow \mu \nu) / \Gamma(\pi^+ \rightarrow \mu \nu) = \tan^2 \theta M_K (1 - M_\mu^2 / M_K^2)^2 / M_\pi (1 - M_\mu^2 / M_\pi^2)^2. \quad (3)$$

From the experimental data, we then get^{5,6}

$$\theta = 0.257. \quad (4)$$

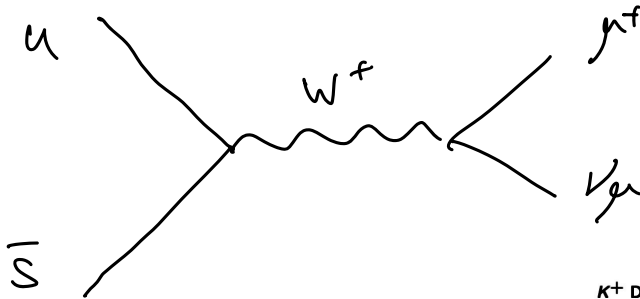
For an independent determination of θ , let us consider $K^+ \rightarrow \pi^0 + e^+ + \nu$. The matrix element for this process can be connected to that for $\pi^+ \rightarrow \pi^0 + e^+ + \nu$, known from the conserved vector-current hypothesis (2nd assumption). From the rate⁶ for $K^+ \rightarrow \pi^0 + e^+ + \nu$, we get

$$\theta = 0.26. \quad (5)$$

The two determinations coincide within experimental errors; in the following we use $\theta = 0.26$.

We go now to the leptonic decays of the baryons, of the type $A \rightarrow B + e + \nu$. The matrix element of

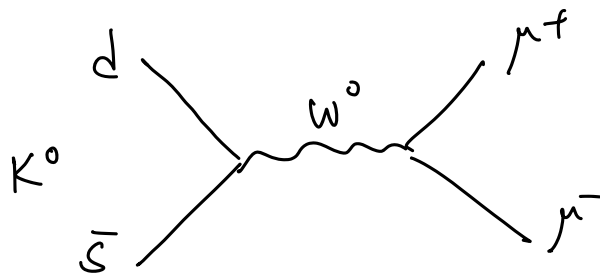
$$K^+ \rightarrow \mu^+ \nu_\mu$$



$$BF = 66\%$$

$$Q = m_K - m_\mu = 394 \text{ MeV}$$

K^+ DECAY MODES	Fraction (Γ_i/Γ)	Scale factor/ p Confidence level (MeV/c)	
Leptonic and semileptonic modes			
$e^+ \nu_e$	(1.582±0.007) $\times 10^{-5}$		247
$\mu^+ \nu_\mu$	(63.56 ±0.11) %	S=1.2	236
$\pi^0 e^+ \nu_e$	(5.07 ±0.04) %	S=2.1	228
Called K^+_{e3} .			
$\pi^0 \mu^+ \nu_\mu$	(3.352±0.033) %	S=1.9	215
Called $K^+_{\mu 3}$.			
$\pi^0 \pi^0 e^+ \nu_e$	(2.55 ±0.04) $\times 10^{-5}$	S=1.1	206
$\pi^+ \pi^- e^+ \nu_e$	(4.247±0.024) $\times 10^{-5}$		203
$\pi^+ \pi^- \mu^+ \nu_\mu$	(1.4 ±0.9) $\times 10^{-5}$		151
$\pi^0 \pi^0 \pi^0 e^+ \nu_e$	< 3.5 $\times 10^{-6}$	CL=90%	135
Hadronic modes			
$\pi^+ \pi^0$	(20.67 ±0.08) %	S=1.2	205
$\pi^+ \pi^0 \pi^0$	(1.760±0.023) %	S=1.1	133
$\pi^+ \pi^+ \pi^-$	(5.583±0.024) %		125



Charge conjugation $\times$ Parity (CP) or Lepton Family number (LF) violating modes, or $\Delta S = 1$ weak neutral current (S1) modes

	S1		CL
$\Gamma_{23} \mu^+ \mu^-$	$(6.84 \pm 0.11) \times 10^{-9}$		
$\Gamma_{24} e^+ e^-$	$(9 \pm 4) \times 10^{-12}$		
$\Gamma_{25} \pi^+ \pi^- e^+ e^-$	$(3.11 \pm 0.19) \times 10^{-7}$		
$\Gamma_{26} \pi^0 \pi^0 e^+ e^-$	$< 6.6 \times 10^{-9}$		CL=90%
$\Gamma_{27} \pi^0 \pi^0 \mu^+ \mu^-$	$< 9.2 \times 10^{-11}$		CL=90%
$\Gamma_{28} \mu^+ \mu^- e^+ e^-$	$(2.69 \pm 0.27) \times 10^{-9}$		
$\Gamma_{29} e^+ e^- e^+ e^-$	$(3.56 \pm 0.21) \times 10^{-8}$		
$\Gamma_{30} \pi^0 \mu^+ \mu^-$	CP, S1 [e] $< 3.8 \times 10^{-10}$		CL=90%
$\Gamma_{31} \pi^0 e^+ e^-$	CP, S1 [e] $< 2.8 \times 10^{-10}$		CL=90%
$\Gamma_{32} \pi^0 \nu \bar{\nu}$	CP, S1 [f] $< 3.0 \times 10^{-9}$		CL=90%
$\Gamma_{33} \pi^0 \pi^0 \nu \bar{\nu}$	S1 $< 8.1 \times 10^{-7}$		CL=90%
$\Gamma_{34} e^\pm \mu^\mp$	LF [a] $< 4.7 \times 10^{-12}$		CL=90%

$$Q = m_{K^0} - 2m_\mu = 500 - 212 = 288 \text{ MeV}$$

$$P(K^0 \rightarrow \pi^+ \pi^-) \approx 10^{-9} \ll P(K^0 \rightarrow \pi^+ \nu \mu^-)$$

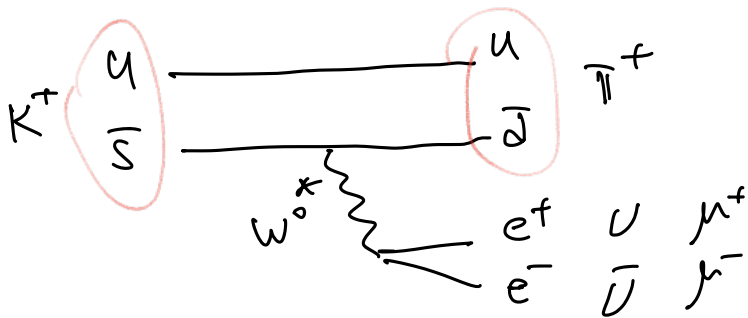
At leading level how to suppress W^0 -mediated process?

$$- g_W^0 < g_W^\pm$$

$$- m_{W^0} \gg m_{W^\pm}$$

W^0 % neutral flavor changing weak current

If W^0 exists $\Rightarrow$ what else should be possible



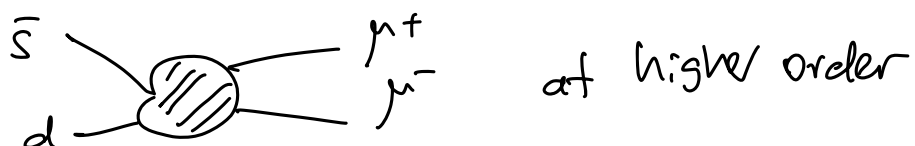
$$m_K - m_\pi = 500 - 140 = 360 \text{ MeV}$$

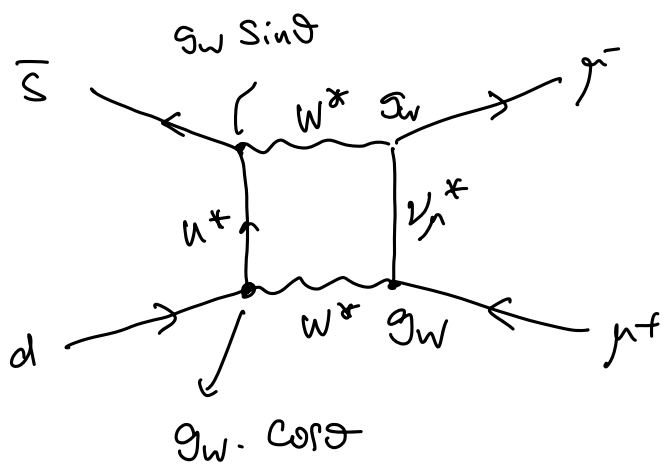
not observed.

$\Rightarrow$ probably W^0 does not exist

$$\Rightarrow \text{how } K^0 \rightarrow \pi^+ \pi^- \approx 10^{-9} \neq \emptyset$$

$\Rightarrow$ look at next order (loops) with $W^\pm$





$$q^+ \rightarrow \frac{q_\mu \gamma^\mu}{q^2}$$

$$\mathcal{M} \sim \underbrace{\frac{g_W^2}{m_W^2}}_{G_F} \underbrace{\frac{g_W^2}{m_W^2}}_{G_F} \sin \theta \cos \theta$$

$$\Gamma(K^0 \rightarrow \pi^+ \pi^-) \propto G_F^4 \sin^2 \theta \cos^2 \theta \gg 10^{-9} \text{ (exp value)}$$

$$\left(\frac{\sin \theta}{\cos \theta}\right)^2 = 0.05 \quad G_F = 1.16 \times 10^{-5} \text{ GeV}^{-2}$$

higher order estimate $\neq$ measured value

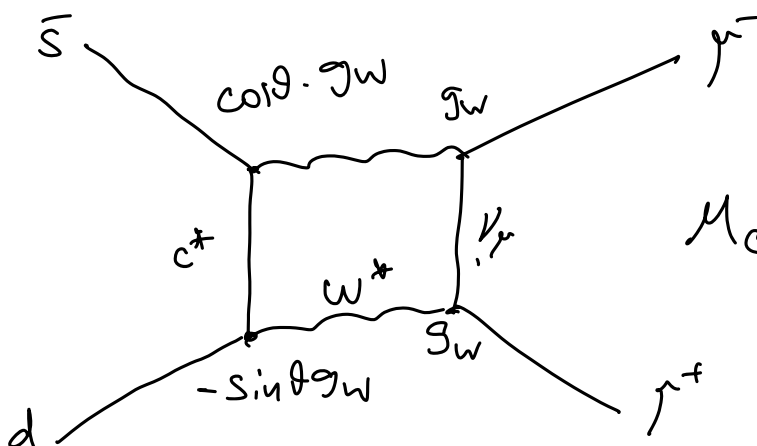
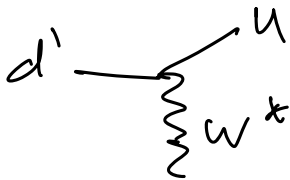
Solution: Glashow-Iliopoulos-Maiani mechanism.
 GM (1970)

$\Rightarrow$ proposed existence of charm quark.
 should have same charge as U-quark.

$$Z_c = +2/3$$

$$d' = d \cdot \cos \theta + s \cdot \sin \theta$$

$$s' = -\sin \theta \cdot d + \cos \theta \cdot s$$



$$\mathcal{M}_c \sim \frac{g_W^2}{m_W^2} \frac{g_W^2}{m_W^2} (-\sin \theta)(\cos \theta)$$

$$\begin{aligned}\mathcal{M}(K^0 \rightarrow \mu^+ \mu^-) &= \mathcal{M}_u + \mathcal{M}_c = \\ &= \left(\frac{g_w^2}{m_w^2}\right) \left(\frac{g_w^2}{m_w^2}\right) (\sin\theta \cos\theta \pi_u - \sin\theta \cos\theta \pi_c)\end{aligned}$$

π_q : propagator of q quark

$$\mathcal{M} \neq 0 \Rightarrow m_c \neq m_u$$

$$\begin{aligned}\Gamma(K^0 \rightarrow \mu^+ \mu^-) \approx 10^{-9} &\Rightarrow \text{Constraints on mass of} \\ &\text{charm quark.} \\ &\Rightarrow m_c \in [1, 3] \text{ GeV}\end{aligned}$$